

5	(a) (i)	V_H depends on angle between (plane of) probe and B -field <i>either</i> V_H max when plane and B -field are normal to each other <i>or</i> V_H zero when plane and B -field are parallel <i>or</i> V_H depends on sine of angle between plane and B -field	B1	
			B1	[2]
	(ii)	1 calculates $V_H r$ at least three times to 1 s.f. constant so valid or approx constant so valid or to 2 s.f., not constant so invalid	M1	
			A1	[2]
		2 straight line passes through origin	B1	[1]
	(b) (i)	e.m.f. induced is proportional / equal to rate of change of (magnetic) flux (linkage) constant field in <u>coil</u> / flux (linkage) of <u>coil</u> does not change	M1	
			A1	
			B1	[3]
	(ii)	e.g. vary current (in wire) / switch current on or off / use a.c. current rotate coil move coil <u>towards</u> / <u>away</u> from wire (1 mark each, max 3)	B3	[3]

6 (a) all four diodes correct to give output, regardless of polarity M1